PG No. 20 D_{3d} $\bar{3}m$ (-3m1 setting) [trigonal]

Table 1: Wyckoff site: `1o`, site symmetry: `-3m.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2a`, site symmetry: `3m.`

No.	position	mapping
1	[0, 0, z]	`[1,2,3,10,11,12]`
2	[0, 0, $-z$]	`[4,5,6,7,8,9]`

Table 3: Wyckoff site: `6b`, site symmetry: `.2.`

No.	position	mapping
1	[x, 0, 0]	`[1,5]`
2	[0, x, 0]	`[2,4]`
3	[$-x$, $-x$, 0]	`[3,6]`
4	[$-x$, 0, 0]	`[7,11]`
5	[0, $-x$, 0]	`[8,10]`
6	[x, x, 0]	`[9,12]`

Table 4: Wyckoff site: `6c`, site symmetry: `.m.`

No.	position	mapping
1	[x, $-x$, z]	`[1,10]`
2	[x, $2x$, z]	`[2,12]`
3	[$-2x$, $-x$, z]	`[3,11]`
4	[$-x$, x, $-z$]	`[4,7]`
5	[$2x$, x, $-z$]	`[5,9]`
6	[$-x$, $-2x$, $-z$]	`[6,8]`

Table 5: Wyckoff site: `12d`, site symmetry: `1`

No.	position	mapping
1	[x, y, z]	`[1]`
2	[$-y$, $x-y$, z]	`[2]`
3	[$-x+y$, $-x$, z]	`[3]`
4	[y, x, $-z$]	`[4]`
5	[$x-y$, $-y$, $-z$]	`[5]`
6	[$-x$, $-x+y$, $-z$]	`[6]`
7	[$-x$, $-y$, $-z$]	`[7]`
8	[y, $-x+y$, $-z$]	`[8]`
9	[$x-y$, x, $-z$]	`[9]`

continued ...

Table 5

No.	position	mapping
10	$[-y,\ -x,\ z]$	`[10]`
11	$[-x+y,\ y,\ z]$	`[11]`
12	$[x,\ x-y,\ z]$	`[12]`